

Binary Quadratic Forms and the Zeros of Epstein's Zeta Function



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In 1859, Riemann investigated $\zeta(s)$ and made the famous hypothesis (RH) that all nontrivial zeros of $\zeta(s)$ must lie on the line $\operatorname{Re}(s) = 1/2$. Since then, many variations of $\zeta(s)$ have been developed. While some of them are believed to satisfy the analog of RH, there are cases where RH is false.

One interesting false case is Epstein's zeta function. In 1936, Davenport and Heilbronn showed that the Epstein's zeta function will have infinitely many nontrivial zeros off the critical line if certain quadratic integer rings fail to be unique factorization domains. In this talk, we will survey the theory of binary quadratic forms and its connection to quadratic fields, which are the key ingredients in the proof of the Davenport-Heilbronn theorem.

This lies at the intersection of analytic number theory, algebraic number theory, and Gauss's theory of binary quadratic forms. At its heart is the remarkable chain of connections

$$\text{quadratic forms} \longleftrightarrow \text{ideal classes}$$

Understanding these relationships sheds light on one of the earliest examples where an analogue of the Riemann Hypothesis fails.

Epstein's Zeta Function

Let

$$Q(x, y) = ax^2 + bxy + cy^2$$

be a positive definite binary quadratic form with discriminant

$$D = b^2 - 4ac < 0.$$

The associated Epstein zeta function is defined by

$$Z_Q(s) = \sum_{(m,n) \neq (0,0)} \frac{1}{Q(m,n)^s}, \quad \Re(s) > 1.$$

This is analogous to the Riemann zeta function

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s},$$

except that one sums over values represented by a quadratic form rather than over the positive integers. For example,

$$Q(x, y) = x^2 + y^2$$

gives rise to

$$Z_Q(s) = \sum_{(m,n) \neq (0,0)} \frac{1}{(m^2 + n^2)^s}.$$

Binary Quadratic Forms

A binary quadratic form is an expression

$$ax^2 + bxy + cy^2.$$

Two forms are considered equivalent if one may be obtained from the other by an integral change of variables

$$(x, y) \mapsto (px + qy, rx + sy), \quad \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in SL_2(\mathbf{Z}).$$

Gauss discovered that the equivalence classes of forms of fixed discriminant D carry the structure of a finite abelian group under composition. The order of this group is called the class number and is denoted $h(D)$.

Quadratic Fields and Class Groups

Given a negative discriminant D , one associates the imaginary quadratic field $K = \mathbf{Q}(\sqrt{D})$. Its ring of integers $\mathcal{O}_K$ need not be a unique factorization domain. A classical example is $\mathbf{Q}(\sqrt{-5})$, where

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

gives two distinct factorizations into irreducibles. The obstruction to unique factorization is measured by the ideal class group $\operatorname{Cl}(K)$, whose order is the class number h_K .

A fundamental theorem of Gauss establishes a bijection

$$\{\text{classes of binary quad. forms of disc. } D\} \longleftrightarrow \operatorname{Cl}(K).$$

Consequently, $h(D) = h_K$.

This correspondence is one of the central bridges between quadratic forms and algebraic number theory.

Dedekind Zeta Functions

For a number field K , the Dedekind zeta function is

$$\zeta_K(s) = \sum_{\mathfrak{a}} \frac{1}{N(\mathfrak{a})^s},$$

where the sum ranges over all nonzero ideals of $\mathcal{O}_K$.

When the class number is 1, one has the factorization

$$\zeta_K(s) = \zeta(s)L(s, \chi_D),$$

where χ_D is the quadratic Dirichlet character attached to K .

Decomposition of Epstein's Zeta Function

A crucial fact is that an Epstein zeta function may be expressed in terms of Hecke L -functions associated with characters of the class group. Schematically,

$$Z_Q(s) = \sum_{\chi} c_{\chi} L(s, \chi),$$

where χ ranges over class group characters.

If the class number equals 1, there is only one character and $Z_Q(s)$ is essentially a single L -function. When the class number exceeds 1, however, several distinct L -functions contribute, and $Z_Q(s)$ becomes a nontrivial linear combination of them.

The Davenport–Heilbronn Theorem

In 1936, Davenport and Heilbronn proved that if the associated quadratic field has nontrivial class group, then the analogue of the Riemann Hypothesis fails for the Epstein zeta function. More precisely, under suitable hypotheses they showed that $Z_Q(s)$ possesses infinitely many zeros satisfying

$$\Re(s) \neq \frac{1}{2}.$$

The key insight is that while a single L -function enjoys an Euler product,

Summary

The central theme of the talk is the correspondence

binary quadratic forms $\longleftrightarrow$ ideal classes in quadratic fields
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and its consequences for zeta functions. When the associated quadratic field has class number greater than one, the corresponding Epstein zeta function becomes a nontrivial linear combination of Hecke L -functions. Davenport and Heilbronn exploited this structure to prove the existence of infinitely many zeros away from the critical line $\Re(s) = 1/2$, providing one of the earliest and most striking failures of a Riemann-Hypothesis-type statement.

$$L(s) = \prod_p (1 - a_p p^{-s})^{-1},$$

a nontrivial linear combination of several L -functions generally does not. The loss of the Euler product permits cancellations that produce zeros away from the critical line.

Why Unique Factorization Matters

The statement in the abstract that the theorem applies when certain quadratic integer rings fail to be unique factorization domains is equivalent to the condition $h_K > 1$. Indeed,

$$\mathcal{O}_K \text{ is a UFD} \iff h_K = 1.$$

Thus one obtains the chain of implications

failure of unique factorization

$$\iff \text{nontrivial class group}$$

$$\iff \text{multiple class-group characters}$$

$$\iff \text{Epstein zeta function is a nontrivial}$$

linear combination of L -functions

$$\iff \text{zeros off the critical line.}$$

This is the conceptual heart of the Davenport–Heilbronn theorem.

Questions to Keep in Mind During the Talk

1. Precise decomposition of $Z_Q(s)$ into Hecke L -functions?
2. Where does the assumption $h_K > 1$ enter the proof?
3. Do we require merely $h_K > 1$, or a stronger hypothesis?
4. Quantify the density of zeros off the critical line?
5. What analogues exist for real quadratic fields?
6. How does the argument compare with modern results on linear combinations of automorphic L -functions?